LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to propulsion, principle of jet propulsion, method of jet engine, bypass engine, single and multi-spool engine, turbo jet, turbo fan, turbo prop, turbo shaft, turbo rocket engines etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to propulsion, principle of jet propulsion, method of jet engine, bypass engine, single and multi-spool engine, turbo jet, turbo fan, turbo prop, turbo shaft, turbo rocket engines etc.
5. Specific Objectives	9.1.1.2 Trainee will be able to explain the principle of operation of gas turbine engine.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine cutaway diagrams; Safety: Do not touch hot components; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propulsion system?
(b) Name one non-jet propulsion method used in aviation.
(c) Define thrust in the context of flight.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Propulsion Systems • Principle of Jet Propulsion • Basic Gas Turbine Method • Turbojet Engine Operation	L & Ds	PPT, IB	15 min
(b)	• Turbofan and Bypass Engine Concepts • Single and Multi-spool Engine Configurations • Turbo Prop and Turbo Shaft Engines • Turbo Rocket Engine Principles	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Engine Types • Application of Engines in Aircraft • Summary of Gas Turbine Cycle • Review of Specific Objective 9.1.1.2	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a bypass engine differs from a turbojet.
(b) Describe the role of the compressor in a multi-spool engine.
(c) Identify which engine type uses a propeller driven by a turbine.

12. Home Assignment and Follow Up	Prepare a brief note comparing the thrust generation method of a Turbojet versus a Turbofan engine for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min